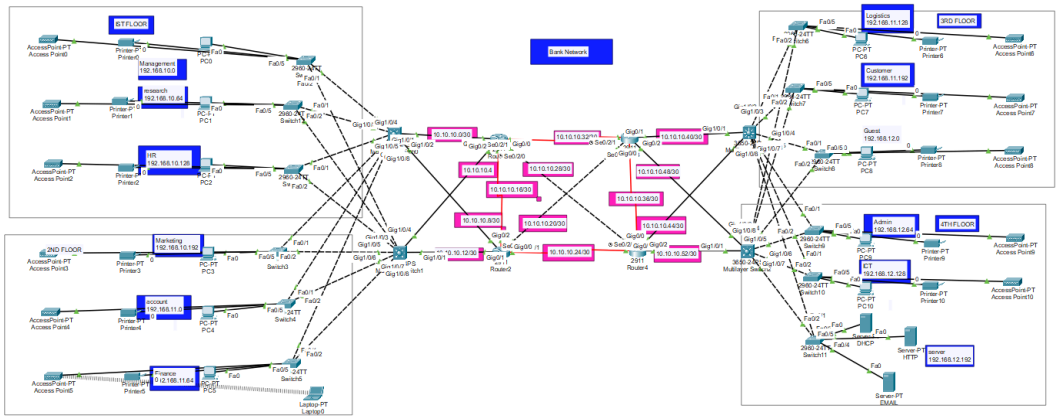
PROJECT DOCUMENTATION: BANK NETWORK DESIGN AND IMPLEMENTATION

Network Topology overview

The topology implements a resilient enterprise network with 4 distribution blocks connected to a core WAN. Each block consists of access switches, end devices, and wireless access points, interconnected via a redundant mesh of routers. The design ensures high availability, scalability, and security with centralized server resources.



Introduction To Devices Used

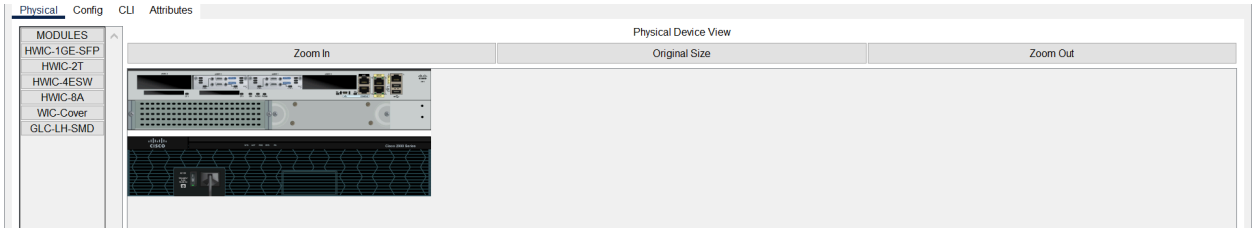
1) Cisco 2911 router
2) 2960 series switches
3) 3650 Multilayer switches
4) Servers
5) End devices(PCs & laptop)
6) Access points
7) Cabling

Configuration of HWIC-2T Modules On All Routers

In Cisco packet tracer, the Cisco 2911 router does not come with serial interfaces by default. To enable WAN connectivity, the HWIC-2T serial module was installed.

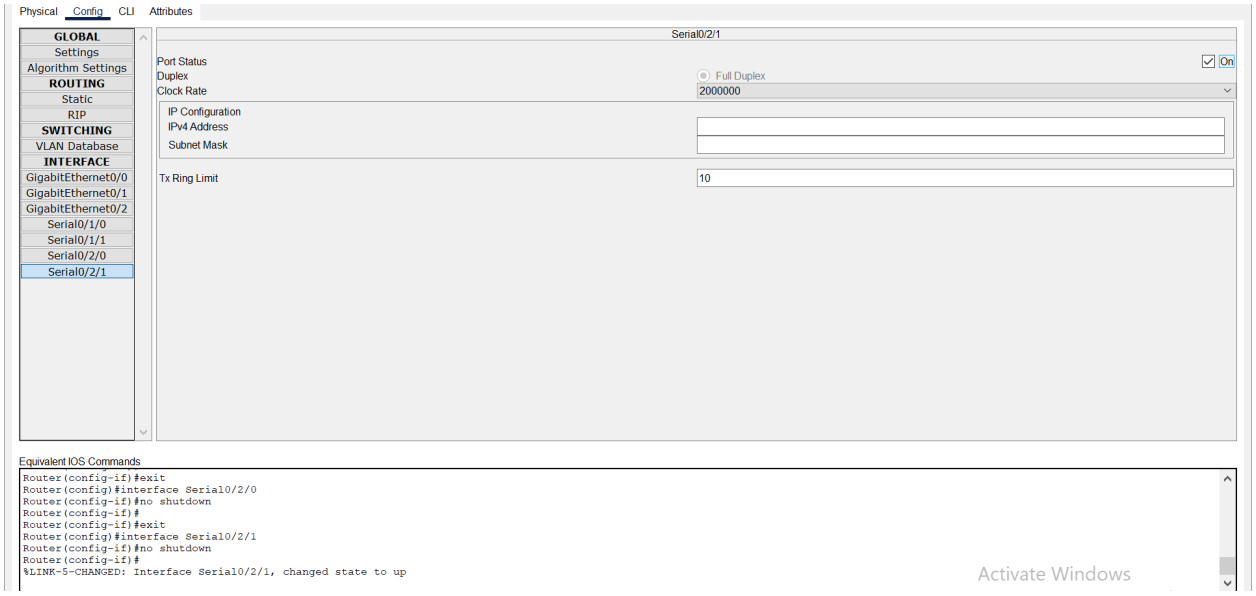
Steps:

1) Click on the 2911 router and go to the physical tab.
2) Power off the router using the power button.
3) Drag the HWIC-2T from the modules list and drop it into an empty slot
4) Power it back on



Enabling Router Interface Using Config

All router interfaces were initially in shutdown state. They were enabled via config. The no shutdown command was used to administratively bring up all LAN and WAN interfaces. The status was verified with show ip interface brief where all active interfaces showed up/up



Basic Settings To All Devices plus SSH

Basic device hardening was performed on all switches and routers. This include setting hostnames, domain names, banners (MOTD), enable secret passwords, and console/vty password. SSH was configured for secure remote management, disabling Telnet to prevent unauthorized access.



VLAN Assignment plus All Access and Trunk

VLANs were created to logically segment the network by department/function. Access points were assigned to end devices based on their respective VLANs. Trunks links are connected between switches and routers using 802.1Q to carry multiple VLANs, ensuring proper traffic isolation.



Switchport security to All 12 switches

Port security was enabled on all 12 access switches to mitigate unauthorized access. This includes limiting the number of MAC addresses per port, learning sticky MAC addresses, and setting violation modes to shutdown, protecting the network from MAC flooding and rogue devices.



Subnetting and IP Addressing

A structured ip addressing scheme was designed using subnetting to efficiently utilize IP addresses and support VLAN segmentation. Each VLAN and point-to-point link was assigned a unique subnet , ensuring logical addressing and future scalability.

FIRST FLOOR

Department	Network address	Subnet mask	Host address range	Broadcast address
Management	192.168.10.0	255.255.255.192/26	192.168.10.1 to 192.168.10.62	192.168.10.63
Research	192.168.10.64	255.255.255.192/26	192.168.10.65 to 192.168.10.126	192.168.10.127
HR	192.168.10.128	255.255.255.192/26	192.168.10.129 to 192.168.10.190	192.168.10.197

SECOND FLOOR

Department	Network Address	Subnet mask	Host address range	broadcast
Marketing	192.168.10.192	255.255.255.192/26	192.168.10.193 to 192.168.10.254	192.168.10.255
Accounts	192.168.11.0	255.255.255.192/26	192.168.11.1 to 192.168.11.62	192.168.11.63
Finance	192.168.11.64	255.255.255.192/26	192.168.11.65 to 192.168.11.126	192.168.11.127

THIRD FLOOR

Department	Network address	Subnet mask	Host address range	Broad cast
Logistics	192.168.11.128	255.255.255.192/26	192.168.11.129 to 192.168.11.190	192.168.11.191
Customer	192.168.11.192	255.255.255.192/26	192.168.11.193 to 192.168.11.254	192.168.11.255
Guest	192.168.12.0	255.255.255.192/26	192.168.12.1 to 192.168.12.62	192.168.12.63

FOURTH FLOOR

Department	Network address	Subnet mask	Host address range	broadcast
Admin	192.168.12.64	255.255.255.192/26	192.168.12.65 to 192.168.12.126	192.168.12.127
ICT	192.168.12.128	255.255.255.192/26	192.168.12.129 to 192.168.12.190	192.168.12.191
Server room	192.168.12.192	255.255.255.19	192.168.12.193	192.168.12.255

		2/26	to 192.168.12. 254	

Base network address: 10.10.10.0

No	Network address	Subnet mask	Broadcast address
1	10.10.10.0	255.255.255.252/30	10.10.10.35
2	10.10.10.4	255.255.255.252/30	10.10.10.39
3	10.10.10.8	255.255.255.252/30	10.10.10.43
4	10.10.10.12	255.255.255.252/30	10.10.10.47
5	10.10.10.16	255.255.255.252/30	10.10.10.51
6	10.10.10.20	255.255.255.252/30	10.10.10.55
7	10.10.10.24	255.255.255.252/30	10.10.10.35
8	10.10.10.28	255.255.255.252/30	10.10.10.39
9	10.10.10.32	255.255.255.252/30	10.10.10.43
10	10.10.10.36	255.255.255.252/30	10.10.10.47
11	10.10.10.40	255.255.255.252/30	10.10.10.51
12	10.10.10.44	255.255.255.252/30	10.10.10.55
13	10.10.10.48	255.255.255.252/30	10.10.10.35
14	10.10.10.52	255.255.255.252/30	10.10.10.39

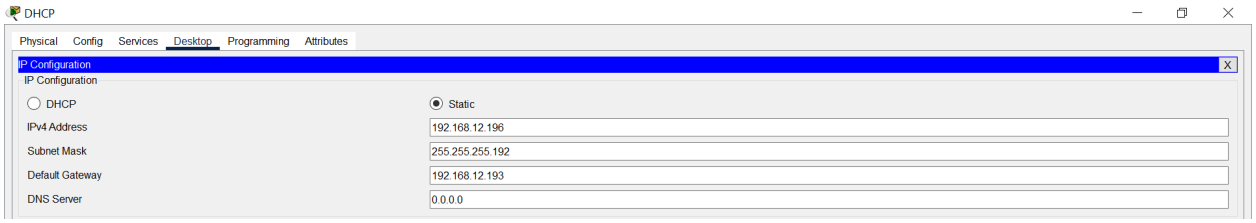
OSPF On Routers and Switches

Open Shortest Path First (OSPF) was configured as a dynamic protocol on all layer 3 devices. This enables dynamic route learning, fast convergence, and load balancing across the redundant WAN links connecting the sites.



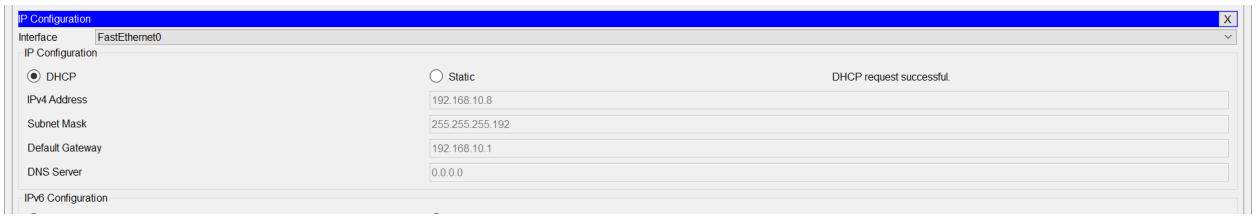
Static IP Addition to Server Room Devices

Servers in the server room (DHCP, EMAIL, HTTP) were assigned static IP addresses to ensure consistent availability and easy identification. This prevents IP conflicts and ensures reliable service delivery.



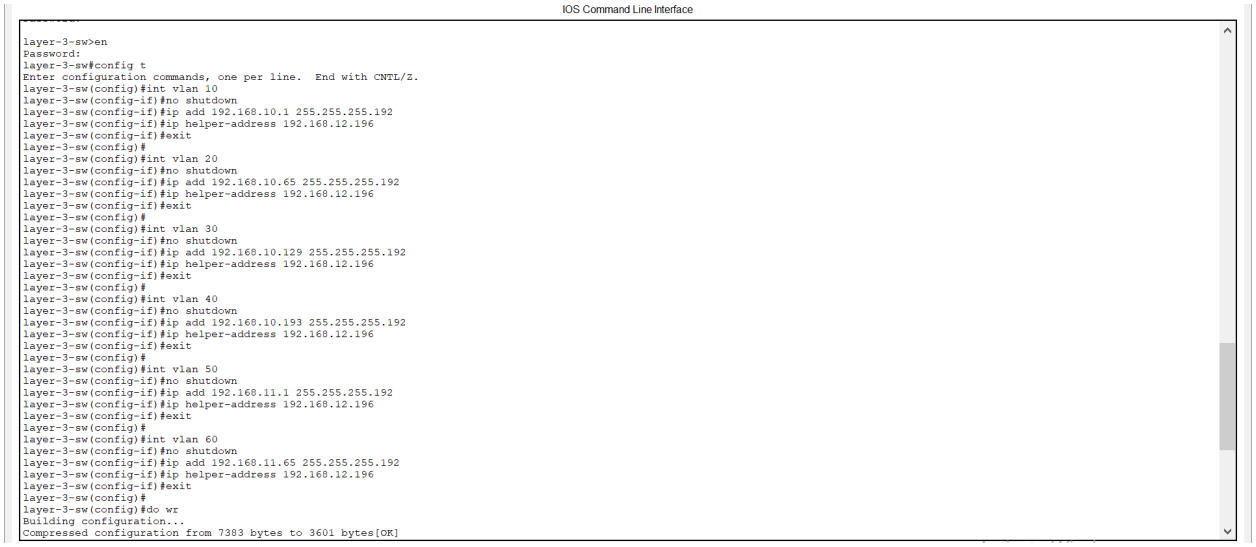
DHCP Server Device Configuration

A centralized DHCP server was configured to automatically assign IP addresses, subnet masks, and default gateways to clients in different VLANs. This reduces manual configuration and IP management overhead.



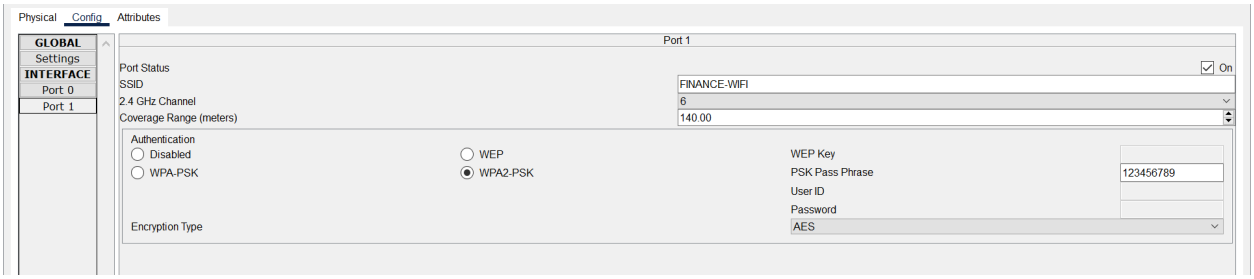
Configure inter-VLAN Routing On Switches Plus IP DHCP Helper Address

Inter-VLAN routing was configured on the multilayer switches to allow communication between different VLANs. IP helper-address was configured to forward DHCP-broadcast requests from clients to the central DHCP server located in a different VLAN.



Access point configuration

The wireless access point was configured and integrated into the appropriate VLANs to provide wireless connectivity for mobile users. WPA2 security was configured with SSID and password for secure wireless access.



Testing & Pinging

Comprehensive connectivity testing was performed to verify the implementation. This includes pinging between VLANs, across sites, to server, and testing DHCP allocation and internet reachability. All successful ping results confirm full network functionality.



Conclusion

This project designed and implemented a secure and fully connected enterprise network in packet tracer using 2911 routers and 2960 switches. All VLANs, routing,DHCP, and security features were successfully configured.